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Lychee cultivar responses to *Phytophthora litchii* highlight candidate genes associated with cell walls and carbohydrate metabolism

Abstract: Lychee cultivars differ in their response to downy blight caused by *Phytophthora litchii*. We used five public sequencing cohorts to identify genes whose infection responses differ between cultivars and assess their consistency across biological contexts. In Guiwei and Yurong1 leaves, 206 of 19445 expressed genes passed statistical, mapping and gene-model checks; 16 passed all internal robustness tests. Twelve of these 16 were prioritised for follow-up, including candidates provisionally associated with carbohydrate metabolism and cell-wall processes. An independent pericarp dataset compared Heiye with Guiwei. LITCHI005518 (putative alpha-xylosidase) and LITCHI001963 met external-support criteria with effects in the same direction as in discovery; five other genes showed significant direction reversals. The externally supported genes were outside the 16-gene internally robust set. A score combining expression of the 206 genes responded in opposite directions to infection in Feizixiao leaves and fruit, indicating tissue dependence. These findings prioritise carbohydrate- and wall-associated candidates for functional tests in defined cultivars and tissues.

Keywords: oomycete disease; host response; differential expression; transcript usage; reproducible bioinformatics; crop improvement

INTRODUCTION

Lychee (*Litchi chinensis*) downy blight, caused by the oomycete *Phytophthora litchii*, causes field and postharvest losses. Understanding the host processes underlying cultivar differences in disease response could guide resistance breeding. A cultivar-by-infection interaction test assesses whether infection responses differ between cultivars; within-cultivar tests assess responsiveness to infection.

Prior studies described distinct early pericarp responses in susceptible and resistant cultivars (Sun et al. 2019), molecular machinery of the pathogen (Sun et al. 2017), antioxidant and phenolic responses (Yi et al. 2010), and LcCDPK expression under downy mildew stress (Liu et al. 2023). PIAvh202 promotes virulence by destabilising a host ethylene-biosynthesis enzyme (Li et al. 2023). Public RNA-sequencing datasets allow us to assess whether cultivar-dependent infection responses also occur across different cultivars, tissues and sampling times.

An exploratory reanalysis of GSE201243 compared mock-inoculated and infected Guiwei and Yurong1 leaves at 24 h, identifying 18 defence-related candidates and promoter hypotheses. We then registered a separate genome-wide secondary analysis, fixing cohort roles, models, thresholds and external tests before its outcomes were computed. We tested 19445 expressed genes for cultivar-by-infection interactions, checked mapping and gene-model quality, assessed robustness to analytical choices, and evaluated the fixed candidate set in independent pericarp and leaf-fruit datasets. We aimed to identify robust cultivar-dependent responses and assess their scope across biological contexts, while distinguishing expression evidence from functional and regulatory hypotheses.

MATERIAL AND METHODS

Study design and registration. After exploratory analysis of discovery data, protocol v1.0.0 was fixed on 18 August 2026, before outcomes of the registered secondary analysis were computed. The five public cohorts had prespecified roles, including an exploratory role for PRJNA1090613 (Table 1; Figure 1A). The discovery candidate set, signature weights and thresholds were fixed before processing external data. Cross-context support means meeting external criteria in a study with a different tissue, comparator or design. Time-stamped amendments included 20 outcome-blind clarifications, corrections and optimisations and three post-outcome changes to technical quality control or file manifests. The primary external study's published conclusions were already known; delayed count-matrix processing provided a computational safeguard, not experimental blinding.

Data provenance and eligibility. NCBI and GEO metadata were reconciled in a biological-unit registry. Database coverage was reconstructed retrospectively because the historical search log was unavailable. A duplicate super-series record was removed; PRJNA1157370 was excluded for studying *Colletotrichum gloeosporioides* and PRJNA268587/GSE63658 for studying senescence and storage without *P. litchii* inoculation. All five prespecified cohorts remained eligible. Discovery required at least three libraries per cultivar-treatment combination. We treated deposited libraries as independent biological units, although source-tree, pooling, harvest and extraction independence was unreported and could not be established by technical checks.

Preprocessing and quality control. The lychee reference genome assembly (SCAU_Lch_v2.0; Hu et al. 2022) and pathogen reference were combined using distinct contig prefixes so pathogen reads could map to their own reference. Paired reads were checked,

trimmed with fastp (Chen et al. 2018), aligned with STAR two-pass mapping (Dobin et al. 2013), counted with featureCounts (Liao et al. 2014) and independently quantified with Salmon (Patro et al. 2017). Libraries required at least 10 million read pairs after trimming and 40% uniquely aligned reads. GenMap measured exon uniqueness (Pockrandt et al. 2020); observed-mapping checks assessed whether candidate effects depended on ambiguously mapped reads.

Genome-wide discovery. DESeq2 fitted gene counts using cultivar + treatment + cultivar:treatment, with Guiwei and mock as reference levels (Love et al. 2014). The interaction coefficient estimates the Yurong1-minus-Guiwei difference in infection response (infected versus mock). Genes required counts ≥ 10 in ≥ 3 libraries; candidates required genome-wide Benjamini-Hochberg $q < 0.05$ and $|\text{estimated interaction log}_2 \text{ fold change}| \geq \log_2(1.5)$ (Benjamini & Hochberg 1995). The primary rule combined a zero-interaction test with filtering by observed effect size. Post hoc sensitivity analyses tested the composite null $|\text{true effect}| \leq \log_2(1.5)$ and calculated apeglm false-sign-or-small values (Zhu et al. 2019); the candidate set remained fixed. Reported effects are unshrunk maximum-likelihood estimates; shrinkage was used only for signature weights.

Exploratory-candidate audit. Guiwei and Yurong1 were analysed separately with infected-versus-mock DESeq2 models using the same count matrix and 19 445-gene universe, with genome-wide Benjamini-Hochberg adjustment per cultivar. The 18 exploratory candidates underwent the same exon-mappability and gene-model checks as genome-wide candidates. The original exploratory pipeline used TopHat2/Bowtie2 (Kim et al. 2013), HTSeq union-mode counting (Anders et al. 2015), and TBtools and PlantCARE for structural and promoter analyses (Chen et al. 2020; Lescot et al. 2002).

Internal robustness. Candidates were retested with Salmon gene counts, edgeR quasi-likelihood inference (Robinson et al. 2010), a counts-per-million filter and all 12 leave-one-library-out refits. Salmon estimates had to match the discovery direction and differ by $\leq 0.5 \log_2$ units. edgeR used the same design, coefficient and 19 445-gene universe, TMM normalisation, robust dispersion estimation and Benjamini-Hochberg $q < 0.10$ with direction agreement. Leave-one-out analysis required direction agreement in all 12 refits and $q < 0.10$ in ≥ 10 . The primary model was refitted after expression filtering. Passing internal robustness required every applicable check, including observed-mapping sensitivity.

External evaluation and signed signature. PRJNA450886 was fitted with a cultivar $\times$ treatment $\times$ time model, treating time as categorical and 24 h as the reference. The primary

contrast estimated the Heiye-minus-Guiwei difference in infection response at 24 h; 6 h and 48 h were secondary. Tests used Benjamini-Hochberg adjustment across the complete candidate-by-contrast family. External support also required $|\log_2 \text{fold change}| \geq \log_2(1.5)$, a confidence interval excluding zero and agreement with the discovery direction. Significant opposite-direction effects were recorded separately. PRJNA922966 used tissue + treatment + tissue:treatment to assess infection responses across tissues in Feizixiao. A library-level score combined the 206 genes using fixed, shrunken discovery effects as weights.

Pathways and transcript usage. Plant Reactome pathways were mapped to lychee through one-to-one *Oryza* orthologues (Naithani et al. 2024). Discovery used the fixed interaction statistic; robustness combined camera competitive testing (Wu & Smyth 2012), ROAST rotation testing (Wu et al. 2010), leading-edge deletion and an expression- and length-matched null. Salmon transcript abundances were analysed with DRIMSeq and DEXSeq under stageR stage-wise error control (Anders et al. 2012; Nowicka & Robinson 2016; Van den Berge et al. 2017). Event-level reporting required meeting the registered criteria for interpretable gene- and transcript-level error control.

Annotation and supporting evidence. The protein encoded by the longest coding sequence represented each gene; ties were resolved lexicographically. DIAMOND matches to reviewed Swiss-Prot proteins (The UniProt Consortium 2023; Buchfink et al. 2015), InterPro/Pfam domains (Jones et al. 2014) and one-to-one *Oryza* orthology provided separate annotation classes. AME tested 927 JASPAR plant motifs in strand-aware 1-kb and 2-kb promoter windows against 100 expression- and GC-matched backgrounds (McLeay & Bailey 2010; Rauluseviciute et al. 2024). Robustness required meeting the criteria in ≥ 80 backgrounds for each window; FIMO provided sensitivity testing (Grant et al. 2011). Small-RNA analysis required a reference for the same species in the audited PmiREN bulk archive (Guo et al. 2020), excluding study-derived substitutes.

Controlled promoter audit. A post hoc PlantCARE audit compared background strategies while holding published totals for six element strings and one standard TCA-element variant fixed. Identical scoring and multiple-testing correction were applied to 100 000 simulated sets of eighteen 2-kb sequences at 34% GC and 100 genomic sets matched by expression and promoter-GC quintiles. The simulated rerun closely reproduced the exploratory pattern. With genomic backgrounds, ABRE gained significance, MeJA-responsive, TCA and TC-rich classes retained it, and ARE lost significance. The original observed-motif table was not archived;

counts in canonical promoters reproduced the published total only for ARE. The audit assessed background sensitivity; regulatory function requires separate evidence.

Evidence tiers and reproducibility. Tier A required meeting discovery criteria, passing all internal robustness, mapping and annotation checks, cross-context support and evidence from another data type attributable to the gene. Tier B required meeting discovery criteria and passing all internal robustness checks, with partial or non-testable external evidence, supporting prioritisation for validation. Tier C required a robust, externally supported pathway. Snakemake enforced the analysis sequence and SHA-256 file checks (Mölder et al. 2021). Complete tables, software versions, amendments and figure-generation checks are in the supplementary archive.

Simulation design. Negative-binomial simulations retained fitted gene means, maximum-a-posteriori dispersions, cultivar and treatment main effects, and library size factors from discovery and the 24-h external subset. Balanced positive and negative interactions were injected at 12 absolute log₂ fold changes from 0 to 4, with 100 simulations per effect size. Discovery simulations injected effects into 262 genes, applying genome-wide adjustment over 19 445 genes and the registered effect filter. External simulations injected effects into two of 177 fixed candidates passing count and dispersion checks, also requiring the prespecified direction and a 95% confidence interval excluding zero. Dispersions were treated as known. The curves describe conditional power for the simulated designs, not full-pipeline power. The 24-h external testing family differs from the complete observed candidate-by-contrast family.

RESULTS

Discovery quality and genome-wide interaction. All 12 libraries passed technical quality checks (Figure 1B). Principal component analysis separated samples by cultivar (PC1, 54.2% of variance) and infection status (PC2, 16.1%) (Figure 1C). Of 19 445 expressed genes, 262 met the registered significance and observed-effect criteria. Mappability and gene-model checks excluded 31 and 25 genes, respectively, leaving a fixed set of 206 candidates (Figure 2A,C). Repeating the multiple-testing adjustment among the 13 602 genes that passed quality control retained all 206 candidates and added 12; whether post-test filtering preserves false discovery rate (FDR) control remains unestablished. Tests against the composite null supported 8 of the 262 statistical candidates and 7 of the 206 that passed quality control. The corresponding apeglm false-sign-or-small values were below 0.05 for 149 and 113 genes, respectively. These sensitivity analyses did not alter the prespecified 206-gene candidate set;

they distinguished selection by observed effect size from evidence that the true effect exceeds the threshold.

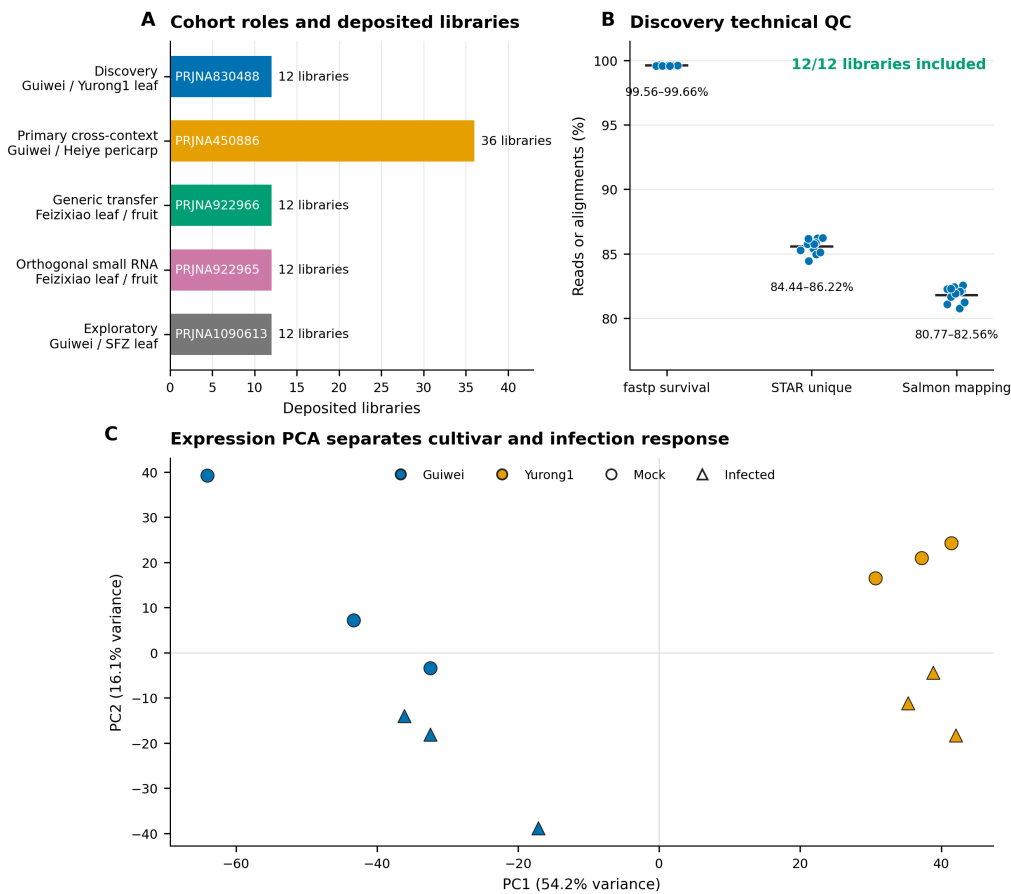


Figure 1. Cohorts and discovery data quality. (A) Five cohorts and their assigned roles. (B) Read retention after filtering, STAR unique alignment and Salmon mapping. (C) Principal component analysis separates samples by cultivar and infection status.

Table 1. Dataset designs and assigned roles.

Accession	Design	Assigned role
PRJNA830488 / GSE201243	Guiwei and Yurong1 leaf; 24 h	Discovery
PRJNA450886	Guiwei and Heiye pericarp; 6, 24, 48 h	Primary external
PRJNA922966 / GSE222651	Feizixiao leaf and fruit; 24 h	Infection response across tissues
PRJNA922965 / GSE222650	Feizixiao small RNA; 24 h	Small-RNA evaluation
PRJNA1090613 / GSE262200	Guiwei and SFZ leaf	Exploratory only

Exploratory candidates and internal robustness. Of the 206 discovery candidates, 18 passed all four expression checks and 16 also passed the observed-mapping checks (Figure 3A). DESeq2 and edgeR effect estimates agreed closely ($r = 0.995$ genome-wide and $r = 0.999$ among candidates), and all candidate effects agreed in direction. edgeR retained 19 genes. The difference in significance between methods remains unresolved, so prioritisation focuses on the

16 genes that passed all internal checks. Separately, among the 18 original exploratory candidates, 13 were significant within Guiwei and 16 within Yurong1 at genome-wide $q < 0.05$; 13 and 15, respectively, also exceeded the observed-effect threshold. Most of these exploratory genes responded to infection within a cultivar, but none met the criterion for a difference in response between cultivars. Two failed the uniform-mappability check.

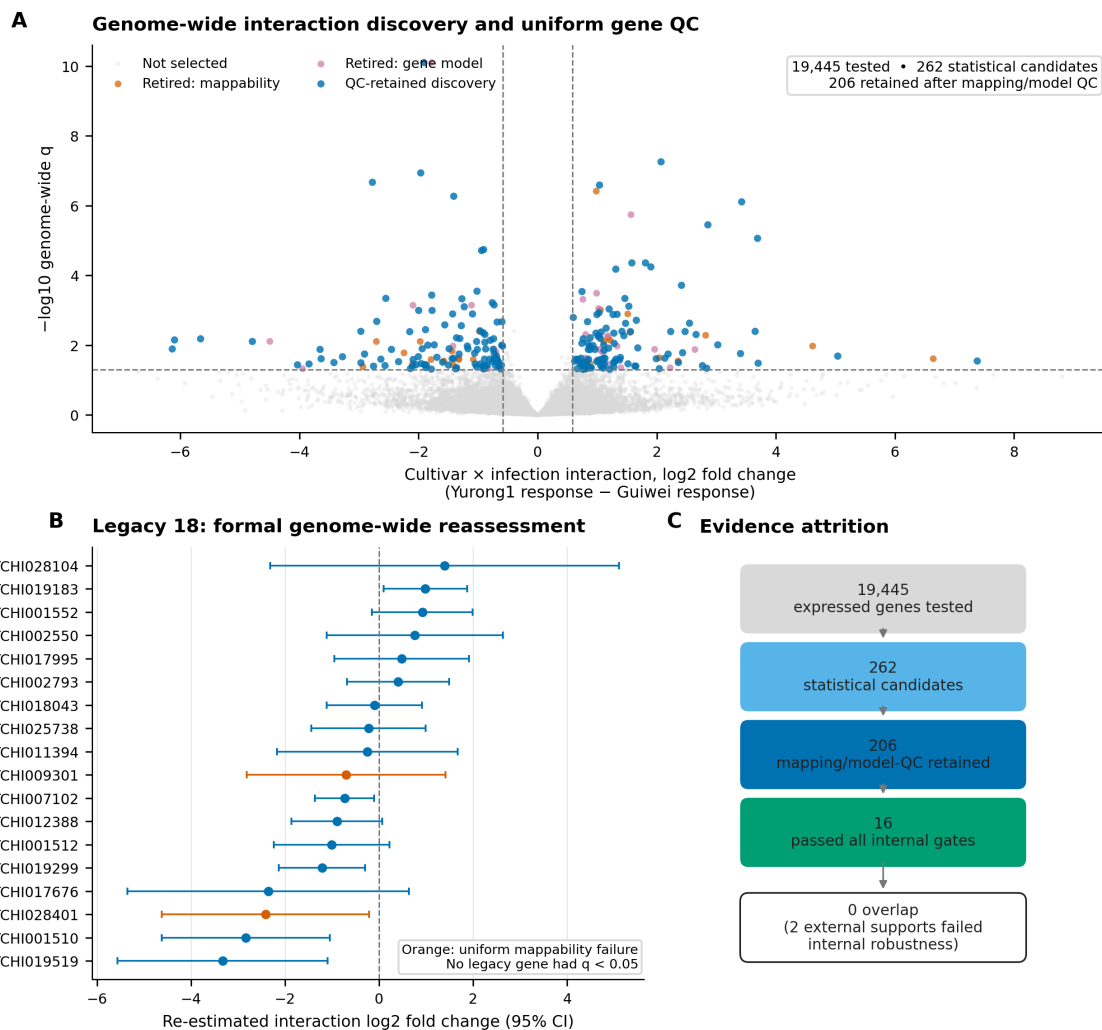


Figure 2. Discovery analysis and reassessment of exploratory candidates. (A) Interaction effects across 19 445 genes; 262 passed statistical criteria and 206 passed quality control. (B) Effects for the 18 original exploratory candidates. (C) Candidates retained through internal robustness checks.

Cross-context outcomes. In the primary external evaluation at 24 h in pericarp, 184 of the 206 candidates were measurable. LITCHI005518 and LITCHI001963 met all external criteria, with effects in the same direction as discovery; five genes met the magnitude and significance criteria in the opposite direction (Table 2; Figure 3C). The remaining candidates fell below the thresholds or could not be tested. Across measurable genes, discovery and external effects had a Pearson r of -0.05 (95% confidence interval -0.19 to 0.10) and a Spearman ρ of -0.04; 95 of 184 agreed in direction. Neither externally supported gene belonged to the 16-gene internally

robust set (Figure 3B). These comparisons assess whether findings extend across studies in which tissue, cultivar comparator and study conditions differ together.

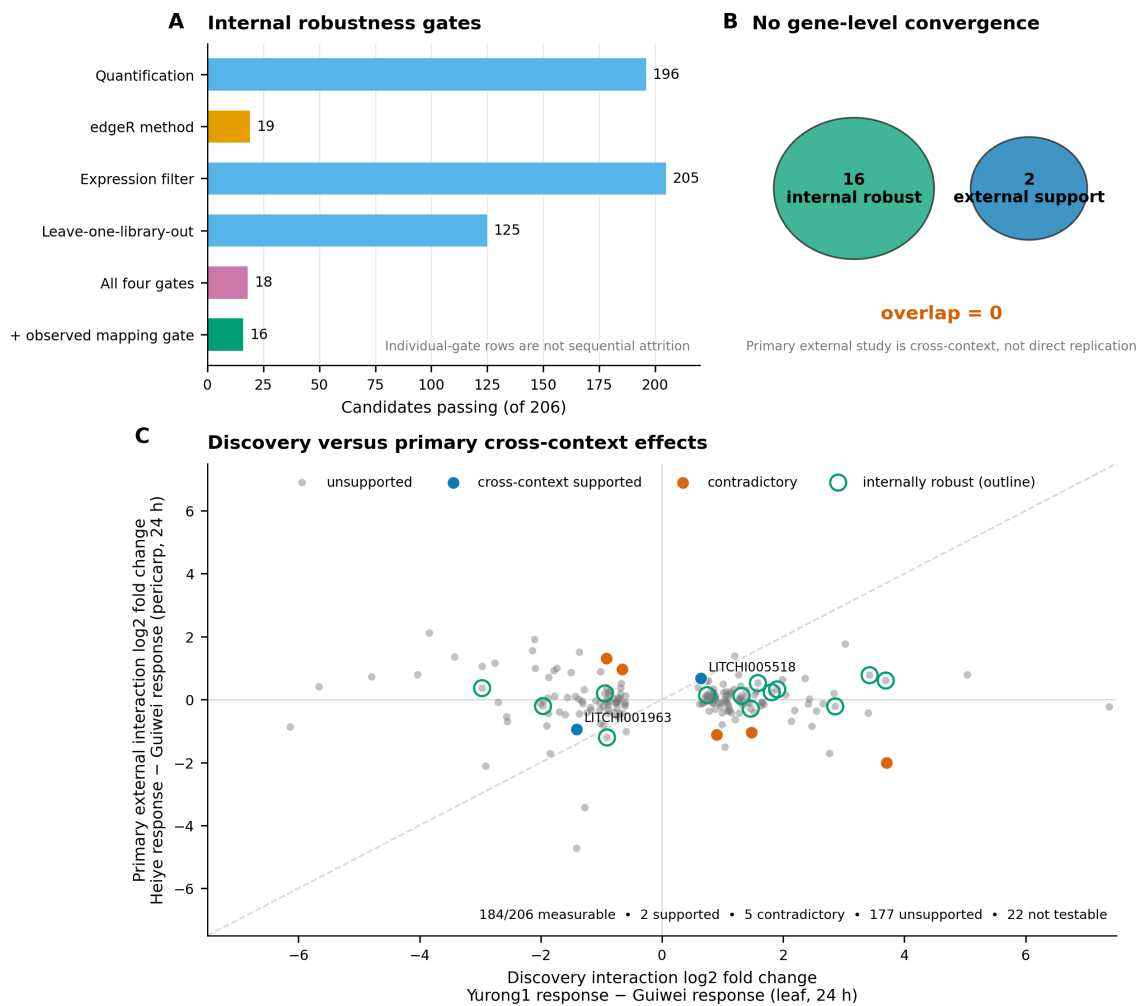


Figure 3. Internal robustness and primary external evaluation. (A) Candidates retained after each robustness check. (B) The two externally supported genes are outside the 16-gene internally robust set. (C) Discovery versus 24-h pericarp effects; the diagonal is a visual reference. Support refers only to the external criteria.

Table 2. Genes with supported or opposite-direction effects in the primary external evaluation. Effects are cultivar-by-infection interaction log2 fold changes, with 95% confidence intervals for external estimates. Annotations are provisional. Both supported genes met the external criteria but did not pass all internal robustness checks.

Gene	Annotation	Discovery log2FC	Discovery q	External log2FC (95% CI)	External q	Outcome
LITCHI005518	Alpha-xylosidase 1	+0.64	0.025	+0.67 (+0.32 to +1.02)	0.007	Supported
LITCHI001963	Uncharacterised; 1:1 Oryza orthologue	-1.41	0.024	-0.94 (-1.43 to -0.46)	0.007	Supported
LITCHI030761	Caffeoylshikimate esterase	-0.92	0.040	+1.31 (+0.78 to +1.85)	0.0002	Reversed
LITCHI029534	Thiamine thiazole synthase 2	+1.48	0.002	-1.04 (-1.61 to -0.48)	0.010	Reversed
LITCHI021860	Phosphatidylinositol 4-kinase gamma 2	-0.65	0.032	+0.97 (+0.39 to +1.54)	0.029	Reversed
LITCHI008313	Beta-glucosidase 44	+0.90	0.045	-1.12 (-1.81 to -0.44)	0.034	Reversed
LITCHI001700	Fe(II)/2-OG-dependent dioxygenase 4	+3.70	0.033	-2.00 (-3.24 to -0.77)	0.035	Reversed

Signatures, pathways and transcript usage. The fixed 206-gene score increased with infection in Feizixiao leaves (0.107, $q = 0.0015$) and decreased in fruit (-0.181, $q < 0.001$), with a tissue-by-infection interaction of 0.289 ($q < 0.001$). In pericarp, cultivar-by-infection effects on the score were 0.109 at 24 h (95% confidence interval -0.079 to 0.297, $q = 0.349$) and -0.240 at 48 h (95% confidence interval -0.428 to -0.052, $q = 0.043$). A direct comparison between times is needed to establish a temporal difference (Figure 4B). The circadian rhythm pathway alone passed all internal robustness checks; none of the six discovery pathways met the external-support criteria (Figure 4A). After the criteria for event-level reporting were met, transcript-usage analysis identified 225 events across 152 genes among 15790 transcripts. In the primary external cohort, 125 events were below threshold and 100 could not be tested. Incompatible annotations prevented testing in the Feizixiao dataset (Figure 5). The exploratory cohort yielded four supported events and five with contradictory evidence. These results were interpreted descriptively because sampling time and resistance status were unresolved, and the cohort's score estimate was excluded from evidence tiers. The discovery events provide transcript-level candidates for follow-up.

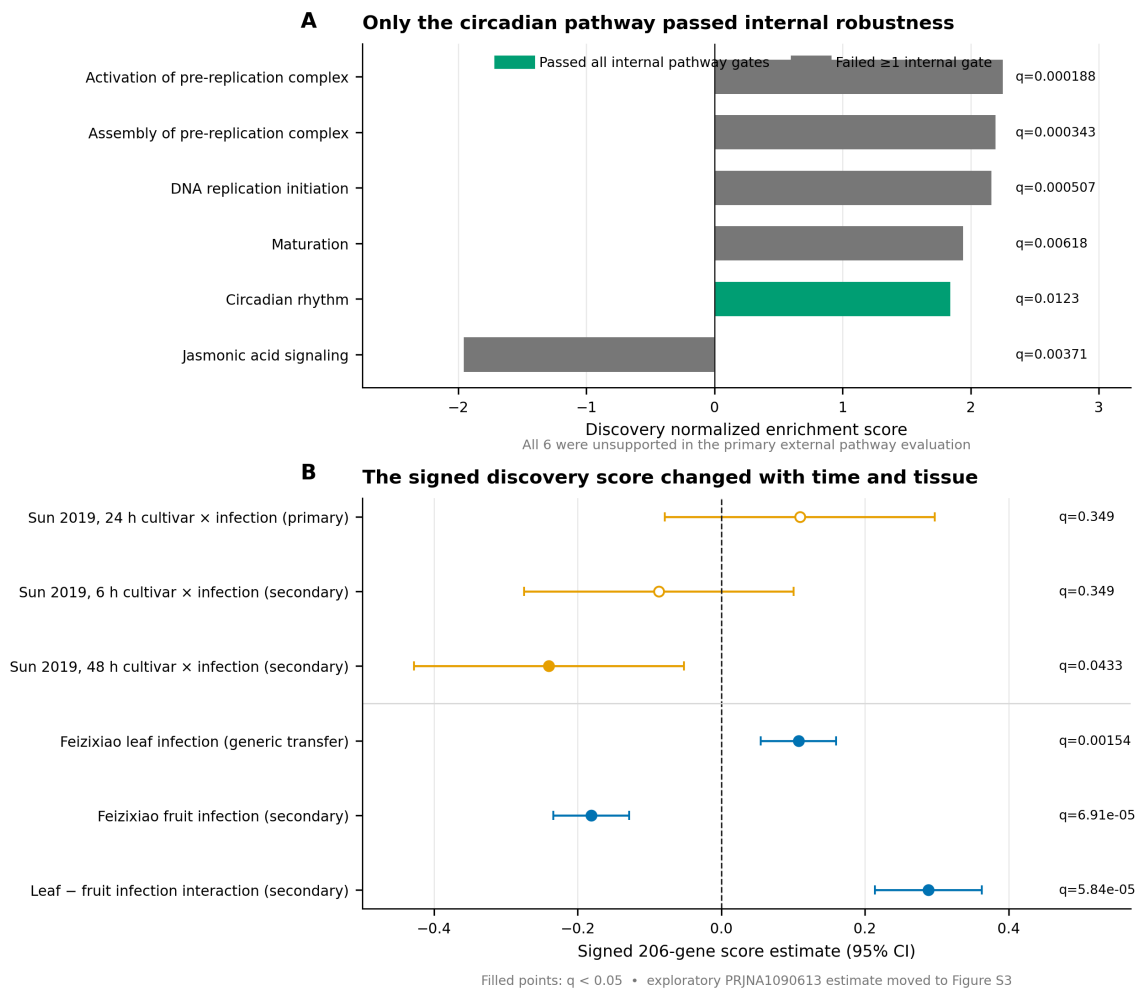


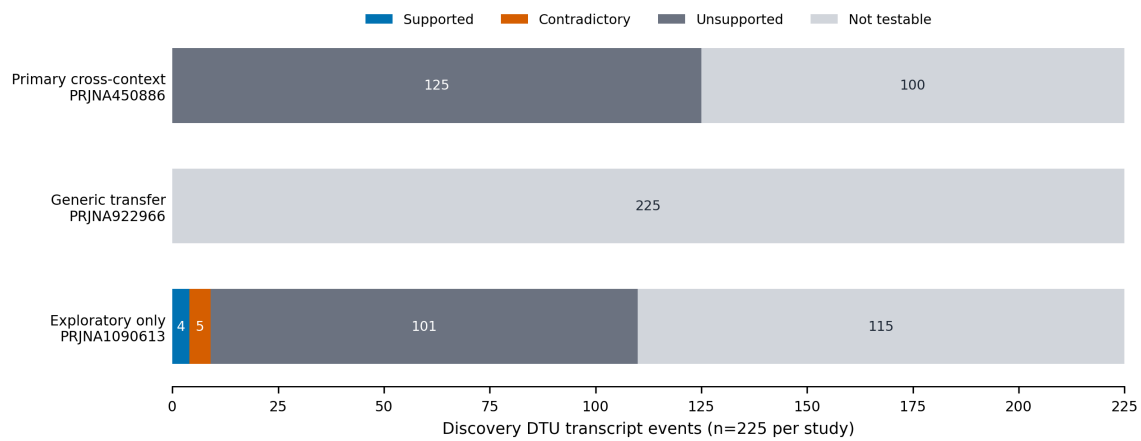
Figure 4. Pathways and gene-expression scores. (A) Circadian rhythm alone passed all internal robustness checks; none of the six discovery pathways met external-support criteria. (B) The fixed 206-gene score across six non-exploratory external contrasts; filled points denote $q < 0.05$. Tissue contrasts test infection responses within Feizixiao. Temporal differences require direct comparisons between times.

A Conditional DTU discovery



Stage-wise q values and signed usage-interaction differences

B External DTU follow-up remained separated by study role



No primary event was supported; exploratory outcomes remain non-confirmatory.

Figure 5. Transcript usage after the reporting criteria were met. (A) Discovery criteria and 225 events across 152 genes. (B) External evaluation by prespecified study role.

Supporting evidence, tiers and detectable effects. Combining the evidence prioritised 12 Tier B genes, with 191 entries classified as exploratory and 65 as retired from prioritisation. No genes qualified for Tier A and no pathways for Tier C (Figure 6C). All nine Tier B genes measurable externally had effects in the same direction as discovery, although they did not meet all external criteria. Family-level annotations were assigned to 150 candidates. None met all strict annotation requirements: two annotation classes, at least 70% supported coverage and no conflict in protein architecture. No promoter motif met the registered criterion using matched backgrounds. Small-RNA support could not be assessed because PmiREN lacked an exact-species reference for lychee (Figure 6A,B). Conditional simulations estimated 80% detection power at effect sizes of 2.44 log₂ units in discovery and 2.11 in external candidate-family testing.

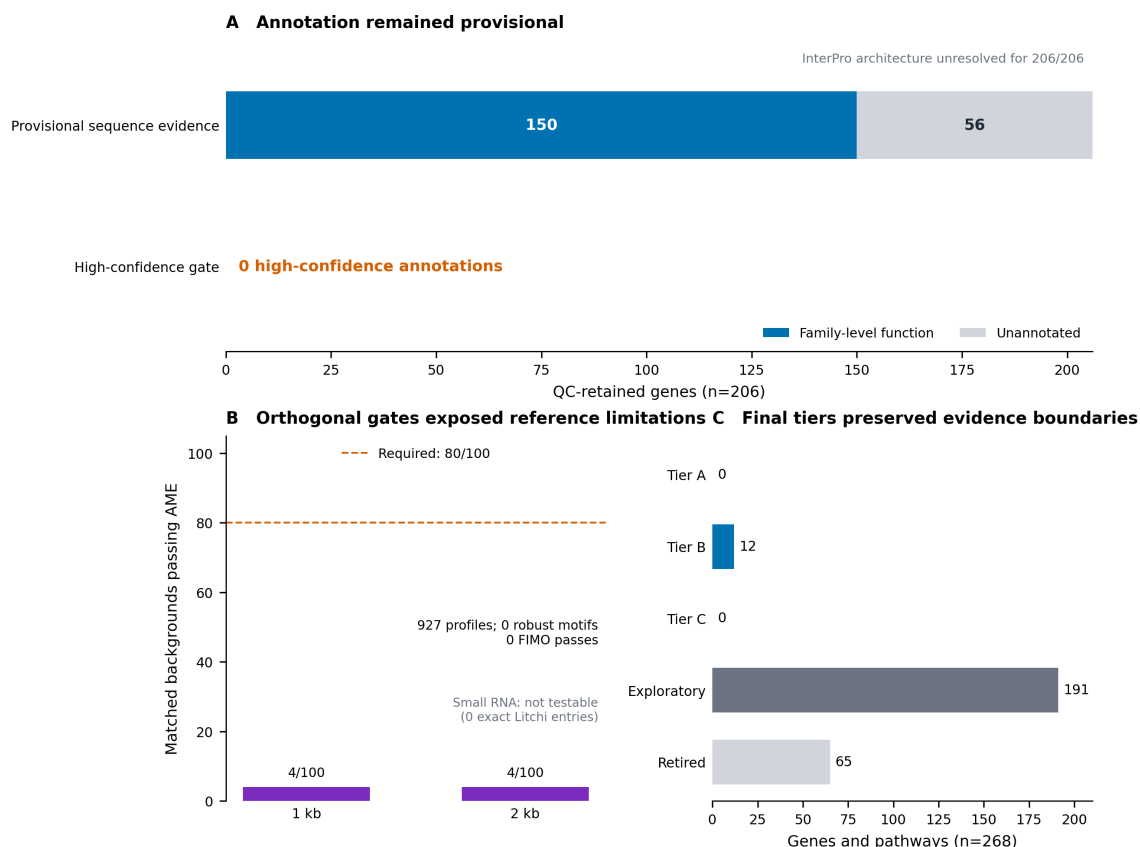


Figure 6. Annotation, additional evidence and evidence tiers. (A) Candidate annotations. (B) Motif robustness and availability of a suitable small-RNA reference. (C) Final evidence tiers.

Tier composition and candidate functions. Provisional annotations for the Tier B genes included an ERD6-like sugar transporter, WAT1-related protein, phosphomannomutase/phosphoglucomutase, starch-branching enzyme, serine carboxypeptidase, DIR1-like lipid-transfer protein, BAG-family regulator, two RNA-metabolism proteins, homeobox-DDT protein and cytochrome P450; one protein was unannotated. These assignments suggest functions for experimental testing and hypotheses for enrichment analysis. Four other internally robust genes, including a putative RPP8-like gene, remained exploratory, with external tests below threshold. Both groups warrant further external testing. The 65 retired entries comprised 56 with mapping or gene-model failures, five with reversed external effects and four with observed-mapping or annotation failures. Technical failures require resolution before further interpretation; reversed effects warrant tests of context dependence.

DISCUSSION

Provisional annotations suggest that genes with cultivar-dependent infection responses may be involved in carbohydrate metabolism and cell-wall remodelling. Twelve Tier B candidates

showed consistent results across internal robustness checks, while two additional genes received support from an independent dataset. Together, these groups identify priorities for experiments testing their roles in lychee disease resistance. No gene met both sets of criteria. Most of the 18 exploratory genes responded to infection within at least one cultivar, but none passed the genome-wide test for differences between cultivar responses. They remain candidates for studying infection responses; their contribution to cultivar differences requires further testing.

The putative phosphomannomutase/phosphoglucomutase, starch-branching enzyme and ERD6-like sugar transporter suggest roles for sugar interconversion, starch metabolism and sugar transport. Their expression changes may reflect carbon use for defence or tissue maintenance, or differences in tissue damage and pathogen growth. Measuring soluble sugars and starch alongside gene expression and disease severity would help relate expression differences to carbohydrate availability. Whether these genes act together in a pathway or alter metabolic flux remains to be tested.

Provisional annotation identifies LITCHI005518 as a possible alpha-xylosidase. In *Arabidopsis*, loss of alpha-xylosidase activity alters xyloglucan composition and growth (Sampedro et al. 2010). The external dataset supported its cultivar-dependent response in the same direction as in discovery. Testing its substrate specificity and effects on wall composition and infection would clarify its possible role in resistance. Sun et al. (2019) reported cultivar differences in infection progression and wall-associated responses in pericarp. Our external evaluation reanalyses their PRJNA450886 dataset and therefore develops the interpretation of the same experiment. The externally supported LITCHI001963 offers a second candidate whose function remains to be characterised.

Arabidopsis DIR1 contributes to systemic acquired resistance (Maldonado et al. 2002), suggesting a function to test for the putative DIR1-like lychee protein. Li et al. (2023) showed that the *P. litchii* effector PIAvh202 destabilises host S-adenosyl-L-methionine synthetase (SAMS) and suppresses ethylene production without changing SAMS transcript abundance. This finding supports combining measurements of transcript abundance, protein abundance and protein activity when evaluating defence candidates.

Discovery compared leaves from Yurong1 and Guiwei; the external study compared pericarp from Heiye and Guiwei. Support for two genes across these comparisons identifies candidates for further study, while limited overall agreement calls for experiments under matched

conditions. The five reversals could reflect context-dependent regulation linked to development or infection progression, or imprecise estimates. Tissue, cultivar pair and study conditions changed together, so their separate effects cannot be identified. In Feizixiao, the discovery-weighted score increased with infection in leaves and decreased in fruit, supporting a tissue-dependent response within that cultivar. A direct comparison of pericarp time points would test whether responses change over time.

The next experiments can address limitations of the public datasets. Each discovery group contained three deposited libraries. Their biological independence was unreported, although statistical inference assumes independence. Using one reference genome may introduce mapping bias, and incomplete metadata limit interpretation. DESeq2 and edgeR estimated similar effects but differed in significance, for reasons that remain unresolved. Support for the full discovery set therefore depends on the statistical method. An external result below threshold may reflect an absent effect or insufficient precision. Pathway and transcript-usage results remain discovery hypotheses because none met the primary external-support criteria. Promoter results remained below threshold, while the lack of an exact-species reference for lychee prevented assessment of small-RNA support.

Independent inoculation experiments in Guiwei and Yurong1 should assess selected internally robust genes alongside LITCHI005518 and LITCHI001963. Matched time courses should measure infected-versus-mock expression in each cultivar, pathogen biomass and disease severity. These measurements would relate expression changes to differences in disease response between cultivars. Matched comparisons across tissues would test whether the responses recur, and quantitative PCR could confirm expression patterns. Altering gene activity and measuring disease and carbohydrate or wall traits would test whether a candidate contributes to resistance or accompanies disease. These experiments would identify candidates for further evaluation as resistance determinants or breeding markers.

CONCLUSION

Provisional gene annotations highlight carbohydrate metabolism and cell-wall processes as priorities for investigating cultivar differences in lychee responses to *P. litchii*. Sixteen genes passed all internal robustness checks, including 12 Tier B candidates, while two additional genes received support from an independent dataset. Opposite infection responses of the combined gene-expression score in Feizixiao leaves and fruit emphasise the importance of tissue context. Functional experiments linking gene activity to pathogen growth and disease

severity can determine whether these candidates contribute to resistance and merit further evaluation for breeding.

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